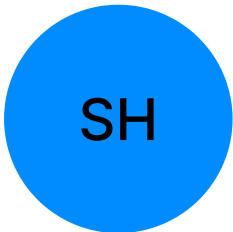


SHANTANU HULE

Python Automation || Devops Engineer || MLOps

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SUMMARY

I am an AI OPS Engineer with a strong foundation in cloud technologies and machine learning. My expertise includes managing and optimizing CI/CD processes, data validation, and deploying complex web applications using modern frameworks. I've contributed to various AI-driven projects, from health diagnostics to cloud-based anomaly detection, demonstrating my ability to elevate operational efficiency. I am passionate about continuous learning and leveraging technology to solve real-world problems

EXPERIENCE

SME Devops

Jio Platforms

📅 03/2025 - Present

Company Description

- Designed and implemented **PowerShell, Bash, and Python automation scripts** for Kubernetes tasks such as deployment backups, ZooKeeper config snapshots, and DR environment config management.
- Developed **alerting scripts** using PowerShell and Python for Kubernetes-based applications to detect failures
- Automated **PVC migration processes** across environments, ensuring zero-downtime transitions and consistent storage configurations.
- Performed **UAT, production, and DR deployments**, including config updates, CRP validations, and rollout monitoring via Azure Pipelines.
- Created a **monitoring microservice** to execute database queries and analyze logs for early anomaly detection in application behavior.
- Integrated **Azure DevOps Pipelines** for config deployments, code changes, and environment-specific automation workflows.
- Analyzed failures in the **Azure Portal Failure section**, traced errors to Kubernetes logs, debugged issues, and coordinated with development teams for resolution.
- Nginx Load Counter Prediction & Auto-Scaling Trigger**: set up a prediction-based scaling mechanism instead of CPU-based scaling, leading to 20% more efficient resource usage.
- Log-Based Anomaly Detection for API Errors**: integrated anomaly detection into our logging pipeline, enabling proactive incident alerts before customer impact
- Container Resource Usage Forecasting: Instead of reacting to resource saturation, we implemented predictive scaling based on ML forecasts

KEY ACHIEVEMENTS



OpenShift Deployment

Successfully deployed a web application on an OpenShift cluster, thoroughly optimized for performance and security



AI Application Development

Created advanced web applications for various AI projects, showcasing expertise in innovative tech solutions

SKILLS

Python AWS Azure Django

Flask Selenium SQL CICD

Data Science ML AI

MLOPS MLflow FastAPI

Docker Kubernetes Helm

Jenkins Prometheus Evidently

Grafana

Devops Git Openshift

Deployment Change management

CICD Sanity Testing Linux

Bash

EXPERIENCE

AI OPS Engineer

Jio Platforms

📅 12/2023 - 03/2025

Company Description

- Developed and deployed AI-driven automation solutions for monitoring and optimizing IT operations.
- Built and maintained automation frameworks using Python, Django, FastAPI, and Selenium.
- Designed and implemented Kubernetes-based deployments for scalable applications, managing ingress configurations, services, and load balancing.
- Created API flow mapping tools using Django (backend) and React (frontend) for better visualization of application architecture.
- Developed and optimized web automation frameworks using Selenium and Appium for automated testing of web and mobile applications
- Integrated Azure DevOps APIs (v5.1) for CI/CD pipeline automation and workflow optimization.
- Deployed machine learning models and data science pipelines for predictive analysis in system performance monitoring
- Troubleshoot and optimized Nginx, OpenShift routes, and Kubernetes networking for secure and efficient API and application access.
- Worked with MongoDB and SQL databases, ensuring efficient data processing and storage solutions.

Machine Learning Intern

Zerot.ai

📅 08/2022 - 03/2023

A tech startup focused on AI and machine learning solutions

- Conducted time series analysis for oil well performance tracking
- Performed anomaly detection for Google Cloud data, identifying behavioral anomalies across multiple cloud services
- Engaged in feature engineering using GCP virtual machine metrics
- Developed ML models and Azure functions for IoT sensor performance evaluation

Artificial Intelligence Intern

Nothink Technologies

📅 04/2022 - 08/2022

Company specializing in data-driven marketing solutions

- Utilized data analysis techniques for Facebook campaigns, optimizing targeting and campaign performance
- Conducted time series analysis on various platforms, including Google, LinkedIn, and Facebook

Artificial Intelligence Intern

Rainet Technologies, Delhi

📅 08/2021 - 03/2022 📍 Delhi

Technological solutions in the AI space

- Performed data analysis for a travel agency, predicting customer claims using machine learning
- Contributed to research paper writing for a medical image classification project
- Applied natural language processing using transformers for feedback classification
- Developed a system to predict customer active and inactive status for a bank

PROJECTS

Healthcare AI Projects

Personal projects focusing on health-related AI applications

- Developed a brain tumor detection model using the darknet framework to predict tumor locations
- Created a chest disease classification model using transfer learning, accessible via GitHub
- Built a Django web application that integrates multiple deep learning models for disease detection

EDUCATION

Bachelor of Engineering

Marathwada Mitra Mandals Institute of Technology

07/2019 - 07/2023